Quantitative Analysis of Media Bias and Stock Price Dynamics: The 2020 Shock

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***Abstract*—This paper studies how media reporting about large-cap firms changed around the 2020 recession and whether shifts in news stance predict firm-level stock outcomes. Using a NewsMTSC pipeline to compute daily target-dependent stance from $690{,}586$ relevance-filtered full-text articles (2015–2025) across 26 S&P 500 firms, we align stance with firm-day returns and market controls and test three questions: whether media stance shifted after January 2020 (RQ1), whether firm-idiosyncratic returns shifted (RQ2), and whether stance predicts returns in a dynamic sense (RQ3). RQ1 and RQ2 use two-way fixed-effects pre/post panel regressions; RQ3 uses a weekly vector autoregression with Granger-causality tests. We find no significant aggregate level shift in either stance or returns once common daily and monthly shocks are absorbed, and a dynamic stance–return link that is firm-specific rather than a market-wide regularity. We document the data pipeline, relevance filtering, stance aggregation, and the robustness checks used throughout.**

I. Introduction

The 2020 recession is a natural breakpoint for studying the interaction between media narratives and financial markets. COVID-era coverage expanded in volume and intensity, and a growing body of evidence documents a structural change in news-market dynamics during and after the shock. This project asks a simple, high-level question: did the tone of firm-specific media reporting shift after 2020, and if it did, did markets respond differently to that tone?

We approach the problem with a target-dependent stance framework rather than generic sentiment. NewsMTSC allows us to score whether an article is positive or negative toward a specific firm even when multiple firms are mentioned in the same text. That distinction is central to firm-level testing: a general sentiment score cannot distinguish positive coverage of one firm from negative coverage of another in the same article. Target-dependent stance makes it feasible to construct a daily, firm-level media stance signal and evaluate how it behaves across time and shocks.

The study is grounded in three observations. First, the 2020 shock is widely documented as a period in which media coverage and market behavior changed in a persistent way. Second, the data pipeline needs to be precise: naive keyword search introduces many false positives and inflates noise in the stance signal. Third, if stance is to be linked to returns in a credible way, the design must rule out coverage volume effects, pre-trend differences, and spurious correlations driven by macro shocks.

We therefore build a two-stage pipeline: (i) relevance classification that filters articles to those that actually report on the target firm, and (ii) NewsMTSC stance scoring with confidence thresholds. With this signal we pose three research questions on structural shifts in stance and returns and the post-shock stance-return linkage. Our empirical strategy combines two-way fixed-effects pre/post panel regressions (RQ1, RQ2) with a weekly vector autoregression and Granger-causality tests (RQ3) to characterize both the level shifts and the dynamic stance-return channel.

The remainder of the paper proceeds as follows. Section II states the research questions and formal hypotheses. Section III describes the data pipeline. Section IV develops the methodology and estimating equations. Section V presents the regression results for RQ1 and RQ2. Section VI presents the VAR and Granger-causality analysis for RQ3. Section VII addresses identification and robustness, and Section VIII concludes.

II. Research Questions and Hypotheses

Three hypotheses organize the empirical work. In each case the null corresponds to no effect; rejection constitutes a positive finding.

RQ1: Stance shift. Did the tone of firm-specific media coverage shift systematically after January 2020?

$$H_0^{(1)} : \beta = 0$$
$$H_1^{(1)} : \beta \neq 0$$

where β is the coefficient on Post_t in the two-way fixed-effects stance regression (Section IV).

RQ2: Returns shift. Did firm-idiosyncratic daily returns shift after January 2020 beyond market co-movement?

$$H_0^{(2)} : \beta = 0$$
$$H_1^{(2)} : \beta \neq 0$$

where β is the coefficient on Post_t in the two-way fixed-effects returns regression.

RQ3: Stance-return predictability. Does past media stance Granger-cause future returns, and did that relationship change post-2020?

$$H_0^{(3)} : a_1^{(2,1)} = \cdots = a_p^{(2,1)} = 0$$
$$H_1^{(3)} : \exists\, \ell \text{ s.t. } a_\ell^{(2,1)} \neq 0$$

where $a_\ell^{(2,1)}$ are the lagged-stance coefficients in the return equation of the VAR.

III. Data and Signal Construction

A. Corpus and Coverage

We begin with a large corpus of scraped full-text news articles (2015–2025) covering large-cap firms that are consistently observable across the window, drawn from the S&P 500. Each record stores the publication, timestamp, URL, title, first paragraph, and full body text. The raw corpus contains 4,706,589 firm-article links (Table I); coverage is highly skewed, with a handful of firms dominating and others sparsely covered.

B. From Corpus to Analysis Sample

The raw corpus is reduced to the analysis sample in three steps: title-level de-duplication, relevance filtering (Section III-C), and target-dependent stance scoring (Section III-D). All downstream analysis runs on the resulting scored table, which contains 690,586 high-confidence firm-article stance observations across 26 firms. Collapsing to the firm–day level yields the 63,200-observation stance panel used for RQ1; aligning stance to firm-day market data yields the 62,667-observation return panel used for RQ2. Table II summarizes the funnel.

Figure 1 shows article coverage per firm in the scored analysis sample. The distribution spans more than four orders of magnitude—from AT&T (139,701 scored articles) down to Eaton (5)—confirming that coverage is concentrated in a subset of heavily reported firms, which the firm fixed effects in Section IV absorb.

C. Relevance Filtering and Annotation

A company name in a headline does not guarantee that the article reports on that company as a corporate entity. “Apple” may refer to a product or appear only in passing, and macro or human-interest stories routinely name a firm without covering it. To remove this noise we keep only articles in which the firm is the primary subject of a material corporate event: earnings, mergers and acquisitions, executive changes, large layoffs, regulatory or legal action, major product launches, or credit and bankruptcy events. The filter has two stages, a title-level pre-filter that requires the company name, not the ticker, to appear in the headline, followed by a supervised classifier that judges materiality. Figure 2 summarizes the full stage.

We manually labeled 750 article titles under this rubric, of which 227 are relevant and 523 are irrelevant, and set aside a separate test set of 190 titles that is never used for training

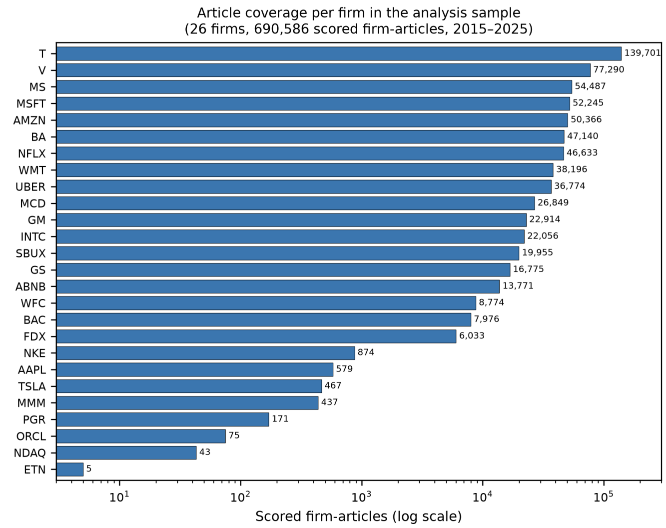

Fig. 1. Scored firm-articles per firm in the analysis sample (26 firms, 690,586 scored firm-articles from `articles_no_title_deduped`, 2015–2025; log scale).

or threshold selection. The classifier is a DeBERTa-v3-base model fine-tuned on titles. Trained on the labeled set alone it reached a precision of 0.65 at a recall of 0.71 (Table III). The binding constraint is the small label budget together with the skew between the labeled relevant rate, near 30%, and the true rate in the corpus, which is closer to 10 to 15%.

To sharpen the decision boundary without further manual labeling, we add 10,025 synthetic titles. These cover the patterns the base model most often confuses, including consumer promotions, industry overviews, immaterial lawsuits, brand comparisons, and articles that name a firm only in passing, together with harder examples of genuine material events phrased in unusual ways. Training on the combined set with focal loss and a fivefold weighting of the labeled examples raises precision from 0.65 to 0.88 while holding recall above 0.85 (Table III).

Table III compares the configurations on the held-out test set. A three-model ensemble with rule-based post-processing reaches the highest precision at 0.94, but we adopt a single consolidated DeBERTa-v3 model: it matches the ensemble on F1, 0.87 against 0.87, using one classifier rather than an ensemble and a separate rule layer, which is simpler to reproduce and to report. At its operating threshold the model attains a precision of 0.88, a recall of 0.86, and an F1 of 0.87; a stricter threshold raises precision to 0.92 at a recall of 0.81 (Figure 3). Applied to the full corpus of 695,731 scored articles it labels 90,579 as relevant, or 13%, consistent with the expected base rate. These relevant articles are the input to the stance stage of Section III-D.

D. Stance Scoring

Stance is computed with the NewsMTSC model for target-dependent stance. For each qualifying article it returns probabilities over positive, negative, and neutral stance toward

TABLE I
YEARLY ARTICLE TOTALS IN THE RAW CORPUS (2015–2025); THESE SUM TO THE 4,706,589 STARTING LINKS.

Year	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Total articles	570,999	625,582	652,741	525,259	559,027	363,962	291,144	260,115	204,281	204,007	449,472

TABLE II
SAMPLE CONSTRUCTION FUNNEL FROM RAW CORPUS TO ESTIMATION PANELS

Stage	Count
Raw corpus (firm-article links, 2015–2025)	4,706,589
Scored firm-articles (relevance-filtered, NewsMTSC)	690,586
Firms with scored coverage	26
RQ1 stance panel (firm–days)	63,200
RQ2 return panel (firm–days)	62,667

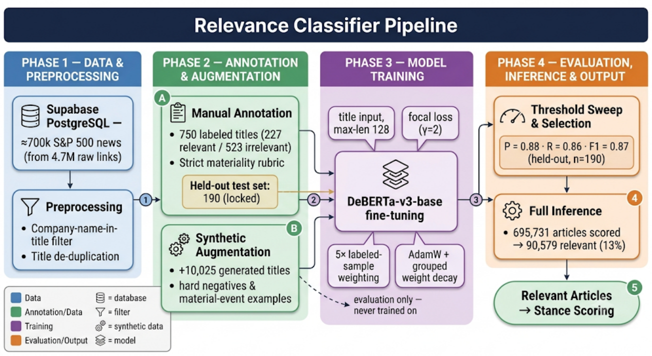

Fig. 2. Relevance classifier pipeline: raw S&P 500 news titles, preprocessing, manual annotation and synthetic augmentation, DeBERTa-v3 fine-tuning, threshold selection on a held-out test set, and full-corpus inference.

TABLE III
RELEVANCE CLASSIFIER PERFORMANCE ON THE HELD-OUT TEST SET ($n = 190$), EACH MODEL AT ITS F1-OPTIMAL THRESHOLD. THE ENSEMBLE IS SHOWN FOR REFERENCE; THE CONSOLIDATED SINGLE MODEL IS ADOPTED.

Training strategy	P	R	F1
Labeled titles only	0.65	0.71	0.68
+ synthetic (broad)	0.78	0.76	0.77
+ synthetic (targeted, focal)	0.86	0.88	0.87
Consolidated single model	**0.88**	**0.86**	**0.87**
Ensemble (reference)	0.94	0.81	0.87

the firm. The baseline specification keeps all scored articles (confidence threshold $\tau = 0$) to maximize coverage; we report sensitivity to higher-confidence subsets ($\tau = 0.6, 0.9$). Daily firm-level stance is the article mean,

$$\text{stance}_{i,t} = \frac{1}{N_{i,t}} \sum_{a \in A_{i,t}} (p_a^+ - p_a^-), \tag{1}$$

where $A_{i,t}$ is the set of scored articles for firm i on day t and $N_{i,t}$ is the count. We also retain article volume as a covariate to separate tone shifts from coverage shifts.

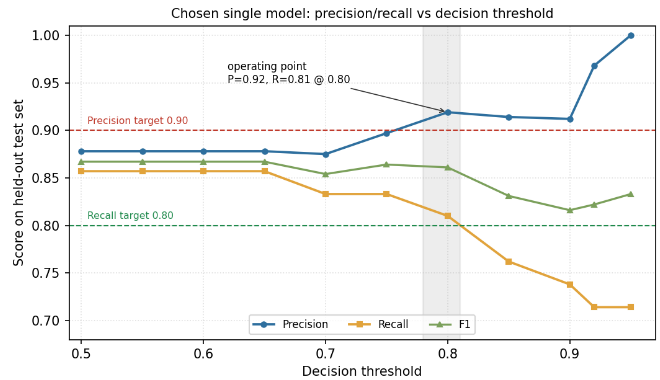

Fig. 3. Precision, recall, and F1 of the chosen single model on the held-out test set as a function of the decision threshold. A threshold near 0.80 satisfies both operating targets, reaching precision 0.92 at recall 0.81, which is why this single classifier is adopted in place of the ensemble.

E. Market Alignment and Controls

The stance series is aligned to firm-day market outcomes: daily returns, the S&P 500 index return, and VIX, drawn from `stock_prices` and the VIX daily series. This alignment yields the panels used in the RQ1–RQ3 specifications.

IV. METHODOLOGY

A. Identification and Panel Design

Both RQ1 and RQ2 use a pre/post design on a firm–day panel. The treatment indicator

$$\text{Post}_t = \mathbf{1}[\, t \geq \text{2020-01-01}\,] \tag{2}$$

partitions the sample into a pre-period (2015–2019) and post-period (2020–2025). Firm fixed effects α_i absorb permanent cross-firm heterogeneity; time fixed effects δ_t absorb common shocks at the day or month level depending on the outcome. The stance outcome is the signed daily mean,

$$\text{bias}_{it} = \frac{1}{|\mathcal{A}_{it}|} \sum_{a \in \mathcal{A}_{it}} (p_a^+ - p_a^-), \tag{3}$$

and the returns outcome is the log return $\text{return}_{it} = \ln P_{it} - \ln P_{i,t-1}$. Both equations are estimated by within-transformation OLS with HC1-robust standard errors; day-clustered standard errors are additionally reported for RQ1.

B. RQ1: Stance Regression

$$\text{bias}_{it} = \alpha_i + \delta_t^{\text{day}} + \beta\, \text{Post}_t + \gamma\, \text{vol}_{it} + \varepsilon_{it} \tag{4}$$

Firm effects α_i absorb each firm's structural coverage level; daily time effects δ_t^{day} absorb market-wide sentiment common to all firms on a given day. The volume control $\text{vol}_{it} = |\mathcal{A}_{it}|$ is required because the mean of a noisy signal mechanically concentrates as more articles are averaged, so volume must be held fixed to isolate a genuine tone shift from a coverage shift. Under $H_0^{(1)}$, $\beta = 0$.

C. RQ2: Returns Regression

$$\text{return}_{it} = \alpha_i + \delta_t^{\text{month}} + \beta\,\text{Post}_t + \gamma_1\,\text{sp500\_ret}_t + \gamma_2\,\text{vix}_t + \varepsilon_{it} \tag{5}$$

Monthly time effects δ_t^{month} absorb common market-level shocks at the monthly frequency, which is the resolution at which macro regime changes operate. The S&P 500 return and VIX are included as daily controls within each month. Under $H_0^{(2)}$, $\beta = 0$.

D. RQ3: VAR and Granger Causality

For the dynamic analysis we estimate a reduced-form bivariate VAR of order p on the endogenous vector $\mathbf{y}_t = (\Delta\text{bias}_t,\, r_t)^\top$:

$$\mathbf{y}_t = \mathbf{c} + \sum_{\ell=1}^{p} \mathbf{A}_\ell\, \mathbf{y}_{t-\ell} + \mathbf{u}_t \tag{6}$$

where Δbias_t is the first-differenced weekly bias index (required for stationarity; see Section VI) and r_t is the weekly log return. The $H_0^{(3)}$ restriction is $a_1^{(2,1)} = \cdots = a_p^{(2,1)} = 0$, tested via a joint Wald F-test with Newey–West HAC standard errors.

V. Results

A. RQ1: No Significant Aggregate Stance Shift

Table IV(a) reports estimates from Eq. (4) on $n = 63{,}200$ firm–day observations (26 firms, 4,018 days). We fail to reject $H_0^{(1)}$: the post-shock coefficient is $\hat{\beta} = +0.0141$ with HC1 SE $= 0.0136$ ($p = 0.30$); day-clustered inference is nearly identical (SE $= 0.0134$, $p = 0.29$), confirming that within-day cross-firm dependence is not driving the result. Article volume is strongly negative (-0.00097^{***}): heavier-coverage firm-days regress toward neutral tone, validating the volume control. The within $R^2 = 0.0017$ reflects the well-known dominance of idiosyncratic article noise at the firm-day level.

The null result is informative. A joint F-test of the 4,017 daily effects is strongly significant ($F = 1.135$, $p = 1.1 \times 10^{-8}$), meaning common daily shocks drive most of the time variation in stance. Once those shocks are absorbed by δ_t^{day}, the static Post step carries almost no residual identifying variation—the day fixed effects effectively span it. We therefore fail to detect a systematic within-firm stance shift beyond what the broader news environment already explains.

B. RQ2: No Significant Firm-Idiosyncratic Return Shift

Table IV(b) reports estimates from Eq. (5) on $n = 62{,}667$ firm–day observations (26 firms; pre-period mean returns tightly clustered, range 0.0019, SD 0.0004). We fail to reject $H_0^{(2)}$: once monthly time effects absorb common market-regime variation, the post-shock coefficient is $\hat{\beta} = -0.0018$ (SE $= 0.0142$, $p = 0.90$). The S&P 500 return regressor is absorbed by the month fixed effects ($\hat{\gamma}_1 \approx 0$, $p = 0.90$), and VIX is similarly uninformative within month ($p = 0.93$). The within $R^2 \approx 0$ follows directly: with the dominant systematic factor (index returns) loaded entirely onto the monthly FE, the regressors have negligible residual explanatory power.

Together, RQ1 and RQ2 establish a clean null baseline: after properly absorbing common daily and monthly shocks, neither media stance nor firm-level returns show a detectable aggregate level shift around January 2020. This motivates the dynamic question in Section VI—whether the *lead–lag relationship* between stance and returns changed post-shock, even if the marginal distributions did not.

VI. Vector Autoregression and Granger Causality Analysis

The regressions in the previous section ask whether the level of each series shifts after 2020. They are silent on the dynamic relationship between media stance and returns: whether past stance helps predict future returns, whether returns feed back into coverage, and whether that lead–lag structure itself changed after the shock. We answer these questions with a vector autoregression (VAR) and Granger-causality framework, supported by an eight-stage data pipeline that constructs a clean, stationarity-validated weekly panel before any dynamic model is fit. We move to a weekly frequency for this analysis because daily firm-level stance is sparse and noisy for several firms, whereas weekly aggregation yields near-complete coverage (median 99.8% of weeks) while retaining enough observations (≈ 522 weeks per firm) for lag-rich models.

A. The VAR Model

For a firm with endogenous vector $\mathbf{y}_t = (\Delta\text{bias}_t,\, r_t)^\top$, where Δbias_t is the first difference of the weekly bias index and r_t is the weekly log return, a reduced-form VAR of order p is

$$\mathbf{y}_t = \mathbf{c} + \sum_{\ell=1}^{p} \mathbf{A}_\ell\, \mathbf{y}_{t-\ell} + \mathbf{u}_t, \tag{7}$$

with coefficient matrices $\mathbf{A}_\ell \in \mathbb{R}^{2\times 2}$ and innovations $\mathbf{u}_t$ having covariance Σ. The off-diagonal entries of the $\mathbf{A}_\ell$ carry the cross-dynamics: a nonzero $(1,2)$ entry means past returns move current stance, and a nonzero $(2,1)$ entry means past stance moves current returns. The VAR is the minimal model that treats both variables as endogenous and lets the data, rather than an a priori causal ordering, reveal the direction of predictability, which is precisely what a media–market feedback question requires.

TABLE IV
REGRESSION RESULTS FOR RQ1 (STANCE SHIFT, EQ. 4) AND RQ2 (RETURNS SHIFT, EQ. 5). BOTH SPECIFICATIONS FAIL TO REJECT THE NULL AT CONVENTIONAL LEVELS ONCE COMMON DAILY (RQ1) AND MONTHLY (RQ2) SHOCKS ARE ABSORBED.

(a) RQ1 — Daily Stance

Firm + day FE, HC1 SE

Variable	Coef.	SE	p
Post ($\hat{\beta}$)	$+0.01413$	0.01362	0.300
Article volume	-0.00097	0.00008	< 0.001
$n = 63{,}200$; 26 firms; 4,018 days			
Within $R^2 = 0.0017$			

(b) RQ2 — Daily Returns

Firm + month FE, HC1 SE

Variable	Coef.	SE	p
Post ($\hat{\beta}$)	-0.00184	0.01424	0.897
S&P 500 return	≈ 0	—	0.902
VIX	$+0.00061$	0.00654	0.925
$n = 62{,}667$; 26 firms			
Within $R^2 \approx 0$			

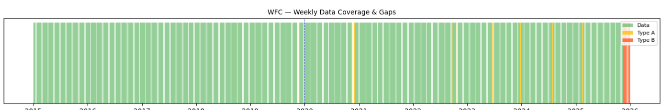

Fig. 4. Weekly data-coverage and gap map for Wells Fargo (WFC), the lowest-coverage retained firm. Short and medium gaps are imputed; the series still clears the $\geq 60\%$ coverage threshold.

B. Stage 1–4: Building the Weekly Panel

The pipeline begins with a *data inventory* (Stage 1) that audits, for every candidate firm, article counts, coverage, and the share of zero-article days, and flags sparse firms ($< 30\%$ trading-day coverage) for exclusion. This audit motivates the move to a weekly frequency and the retention of sixteen well-covered firms. Stage 2 constructs the *weekly bias index* by aggregating signed article stances into ISO weeks and verifies distributional quality (mean, skew, kurtosis, and the incidence of single-article weeks). Stage 3 performs *gap analysis and imputation*: weekly gaps are classified by length, with short Type-A gaps (1–2 weeks) forward-filled and medium Type-B gaps (3–8 weeks) linearly interpolated, while long Type-C gaps are left missing; only 58 observations require imputation and all sixteen firms clear a $\geq 60\%$ coverage threshold with no gap exceeding eight weeks. Crucially, both an imputed and an observed-only series are retained so that every downstream test can be re-run to confirm results are not artifacts of imputation. Figure 4 illustrates the coverage map for the sparsest retained firm.

Stage 4 constructs the *returns series* as weekly log returns from Monday-anchored ISO weeks and aligns them to the bias index. Log returns are used because they are additive across time and approximately symmetric, which suits a linear dynamic model. Quality checks flag extreme weekly moves ($> \pm 15\%$, concentrated in the March 2020 crash window) for later winsorization, and every firm clears the minimum of 100 aligned weeks, yielding a merged panel of 8,090 firm–week rows.

C. Stage 5: Stationarity and Cointegration

A VAR in levels is only valid if the series are stationary or jointly cointegrated; otherwise the regression is spurious.

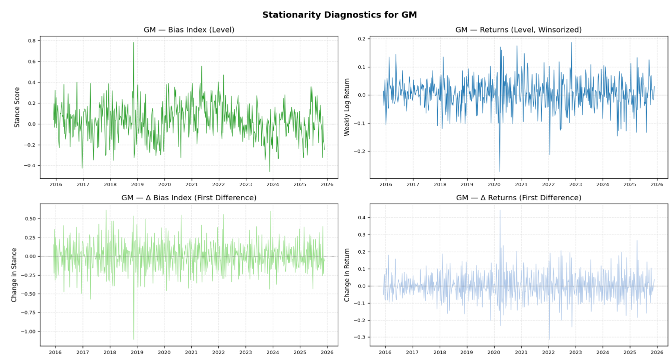

Fig. 5. Stationarity diagnostics for General Motors (GM): the bias index and weekly return in levels (left) and first differences (right), with ADF and KPSS verdicts driving the integration-order classification.

Each series is classified by combining the Augmented Dickey–Fuller (ADF) test, whose null is a unit root, with the KPSS test, whose null is stationarity. A series is called $I(0)$ when ADF rejects ($p < 0.05$) and KPSS does not, $I(1)$ in the opposite case, and "conflicting" (a likely structural break) when both reject. As expected for financial data, weekly log returns are $I(0)$ for essentially every firm, while the bias index is predominantly $I(1)$ or break-driven in levels but $I(0)$ in first differences. This mixed integration order is the reason the VAR is specified on $(\Delta\text{bias}_t, r_t)$ rather than on the levels: differencing the bias index renders it stationary and avoids a spurious levels regression against $I(0)$ returns. A Johansen trace test detects a cointegrating relation only for Wells Fargo (trace $= 340.2$ against a 95% critical value of 15.5), which would otherwise call for a VECM; for the common $(\, I(1), I(0)\,)$ case cointegration is impossible, confirming the differenced specification. Figure 5 shows the level-versus-difference diagnostic for a representative firm.

D. Stage 6: Structural Break Testing

Because the central hypothesis concerns a regime change around 2020, we date breaks empirically rather than imposing the calendar. A Bai–Perron procedure, implemented by dynamic programming for a single change point under an L_2 (mean-shift) cost, locates the largest structural shift in the mean of each weekly series. The estimated bias-index break

Fig. 6. Bai–Perron structural breaks for Visa (V). The return break (2020-03-23) coincides with the COVID crash; the bias break (2020-09-21) lags it.

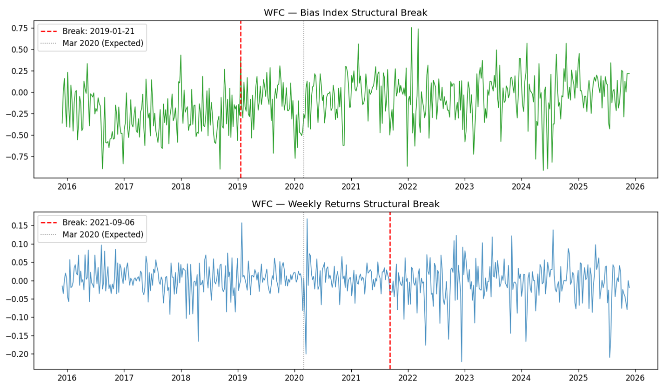

Fig. 7. Bai–Perron structural breaks for Wells Fargo (WFC), the one firm with a cointegrating bias–return relation and significant full-sample causality.

dates serve as the `is_post_break` cut used to split the sample for the subsequent subsample Granger tests, and every firm yields a valid split with at least 60 observations on each side. Figures 6 and 7 show the data-driven breaks for two firms; several cluster near the COVID window while others reflect firm-specific events, which is itself evidence that a single imposed 2020 dummy would be too blunt.

E. Stage 7: VAR Specification

For each firm the VAR order is selected by estimating Eq. (6) for $p = 1, \ldots, 8$ and minimizing the Bayesian Information Criterion (BIC), which penalizes parameters more heavily than AIC and so favors parsimonious, well-identified models; when BIC selects zero lags we impose $p = 1$ to retain a dynamic model. The selected orders range from 1 to 5 (modal value $p = 3$). Three residual diagnostics certify each fit. Stability requires all roots of the companion matrix to lie outside the unit circle, equivalent to $\det(\mathbf{I} - \sum_\ell \mathbf{A}_\ell z^\ell) \neq 0$ for $|z| \leq 1$; every selected model is stable (minimum root > 1.28). The Portmanteau test checks for residual autocorrelation, confirming the chosen lag is adequate for most firms. The Jarque–Bera test rejects normality everywhere, the familiar heavy-tail signature of financial data, which leaves

TABLE V
FIRM-LEVEL GRANGER CAUSALITY, FULL SAMPLE (HAC p-VALUES)

Firm	Lag	Bias→Ret	Ret→Bias
GS	5	0.757	0.012
MS	3	0.058	0.193
WFC	5	0.023	0.307
Other 12 firms: no direction significant at 5%.			

VAR coefficients consistent but motivates robust inference; Engle's ARCH test detects volatility clustering in several return residuals, reinforcing the use of HAC-robust standard errors in the causality stage.

F. Stage 8: Granger Causality

Granger causality operationalizes "does past X help predict Y beyond Y's own past." Stance Granger-causes returns if the lagged stance coefficients are jointly nonzero in the return equation, i.e. $H_0 : a_1^{(2,1)} = \cdots = a_p^{(2,1)} = 0$ is rejected. We test this with a direct OLS regression of each variable on p lags of both variables, using a joint Wald (F) test and Newey–West HAC-robust standard errors (`maxlags`$= p$) to neutralize the heteroscedasticity and autocorrelation found in Stage 7. Table V reports the firm-level full-sample p-values. Causality is sparse in the full sample, as expected if the relationship is regime-dependent: stance Granger-causes returns for Wells Fargo ($p = 0.023$) and weakly for Morgan Stanley ($p = 0.058$), and the reverse direction (returns to bias) is significant for Goldman Sachs ($p = 0.012$).

The more informative test splits each firm at its empirical bias break and re-estimates causality on each side. Several firms that are quiet in the full sample show post-break causality that is absent pre-break: Wells Fargo (bias→returns $p = 0.013$ post versus 0.345 pre) and Morgan Stanley (bias→returns $p = 0.052$ post) are the clearest cases, with feedback (returns→bias) emerging post-break for Boeing, Goldman Sachs, and Morgan Stanley. This pre/post asymmetry is the firm-level analogue of the structural shift the regressions probe: the media–market link is not constant but switches on after the break for a subset of firms.

G. Panel VAR

Firm-by-firm tests are underpowered because each uses only one short series. We therefore pool the cross-section into a panel VAR, which is the design the study most relies on for its dynamic conclusion. Unobserved firm heterogeneity is removed with the Arellano–Bover/Helmert transformation (forward mean-differencing), which subtracts from each observation the mean of all its *future* values within the firm,

$$\tilde{y}_{it} = c_{it}\Big(y_{it} - \tfrac{1}{T_{it}} \textstyle\sum_{s>t} y_{is}\Big), \tag{8}$$

with a scaling factor c_{it} that equalizes the variance of the transformed errors. Unlike ordinary within-demeaning, this transformation preserves the orthogonality between the transformed variables and the lagged regressors, avoiding the look-ahead

TABLE VI
PANEL VAR GRANGER CAUSALITY (15 FIRMS, HELMERT-TRANSFORMED, HAC)

Sample	N	Bias→Ret	Ret→Bias
Full	7493	0.976	0.200
Pre-break	4168	0.245	0.801
Post-break	3265	0.617	0.187

bias (Nickell bias) that contaminates dynamic fixed-effects panels. The endogenous variables are again Δbias and the weekly log return; we additionally partial out the exogenous controls Post$_t$, VIX, the S&P 500 weekly log return, and the article count N_{articles}, so that any detected causality cannot be attributed to market-wide volatility or coverage volume. With $p = 3$ lags and HAC-robust standard errors, the pooled model is estimated by OLS on the transformed, constant-free data, and Granger directions are tested with joint Wald statistics over the full, pre-break, and post-break samples. Table VI reports the results.

Pooling buys power but, even so, neither Granger direction is significant at conventional levels in any subsample once VIX, the index return, and article volume are controlled. The honest reading is that the firm-level post-break signals for Wells Fargo and Morgan Stanley do not aggregate into a uniform panel-wide media-to-returns channel: the dynamic link, where it exists, is firm-specific and heterogeneous rather than a market-wide regularity. This strengthens rather than weakens the contribution, because the panel VAR is exactly the test that would manufacture a spurious aggregate effect if the controls were omitted, and it declines to do so.

H. Limitations of the VAR Pipeline

Several caveats bound the dynamic findings. A weekly frequency, chosen for coverage, may be too coarse to capture the intraday or daily horizon over which news is plausibly priced, so a true fast media-to-price channel could be aliased away by aggregation. The single-break Bai–Perron design dates exactly one change point per series; firms with multiple regime shifts are only partially described, and the break date is estimated with sampling error that the subsample split treats as known. The bivariate VAR omits other drivers of both stance and returns (earnings surprises, analyst actions, sector shocks), so the controls in the panel model mitigate but do not eliminate omitted-variable concerns. Heavy-tailed, volatility-clustered residuals mean small-sample inference leans on the HAC correction and asymptotics rather than exact distributions. Finally, Granger causality is predictive, not structural: a significant test says past stance forecasts returns, not that stance mechanically moves prices, and the firm-specific nature of the surviving signals counsels caution against a universal causal claim.

VII. IDENTIFICATION, VALIDITY, AND ROBUSTNESS

Three threats bear on the level-shift findings. First, macro shocks move both coverage and prices; the day and month fixed effects absorb common shocks, and the S&P 500 return and VIX controls strip the systematic component out of returns. Second, article volume differs widely across firms (Fig. 1); we hold it fixed with the volume covariate, and substituting $\ln(1 + \text{vol})$ for raw volume leaves the RQ1 estimates essentially unchanged. Third, the pre/post design assumes within-firm parallel trends; the flat pre-shock coefficients in the RQ1 monthly event study and the tightly clustered pre-period return means (range 0.0019) are consistent with this premise, though they do not prove it. We further confirm the RQ1 conclusion is robust to the stance confidence threshold ($\tau = 0, 0.6, 0.9$) and certify the estimator with the standard battery (Breusch–Pagan, Jarque–Bera, VIF, Durbin–Watson), which motivates the HC1 and day-clustered inference reported throughout. Two limitations remain: the stance signal inherits measurement noise from the upstream NewsMTSC and relevance stages, which attenuates coefficients toward zero, and a binary post indicator cannot separate the 2020 shock from other contemporaneous regime changes.

VIII. CONCLUSION

We assembled a target-dependent media-stance signal for 26 S&P 500 firms from $690{,}586$ relevance-filtered articles and tested whether the 2020 shock moved stance, returns, and the link between them. The level-shift results are null: once common daily and monthly shocks are absorbed by two-way fixed effects, neither media stance (RQ1, $\beta = +0.014$, $p = 0.30$) nor firm-idiosyncratic returns (RQ2, $\beta = -0.002$, $p = 0.90$) show a significant aggregate shift around January 2020. The dynamic analysis (RQ3) reinforces this reading: a weekly panel VAR finds no market-wide Granger link between stance and returns in any subsample once volatility, the index return, and coverage volume are controlled, with the surviving predictability confined to a few firms (notably Wells Fargo and Morgan Stanley) rather than the panel as a whole. Taken together, the evidence points to a media–market relationship that is firm-specific and heterogeneous rather than a uniform post-2020 regularity—a conclusion that the controlled panel design supports precisely because it declines to manufacture a spurious aggregate effect.

APPENDIX

Table VII reports the decision-threshold sweep for the chosen single model on the held-out test set, which underlies Figure 3. Precision and recall trade off monotonically with the threshold, and a value near 0.80 is the operating point that satisfies both targets.

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TABLE VII
HELD-OUT THRESHOLD SWEEP FOR THE CHOSEN SINGLE MODEL ($n = 190$). THE ADOPTED OPERATING POINT IS SHOWN IN BOLD.

Threshold	P	R	F1
0.50	0.878	0.857	0.867
0.70	0.875	0.833	0.854
0.75	0.897	0.833	0.864
0.80	**0.919**	**0.810**	**0.861**
0.85	0.914	0.762	0.831
0.90	0.912	0.738	0.816
0.92	0.968	0.714	0.822
0.95	1.000	0.714	0.833

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